Kitchen Planner

Proof of concept: an AI-assisted prototype that helps small restaurants decide what to prepare and buy, and shows where food waste comes from.

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Source code (public)	https://github.com/radhahutkey-cpu/food-waste-
Live prototype	https://radhahutkey-cpu.github.io/food-waste-/
Status	Working prototype tested on sample data. Owner interviews not yet done.

The brief (original prompt)

> **Who**
> Independent restaurant owners running 30-60 seat restaurants who manage daily food preparation, purchasing and inventory without a dedicated data analyst.
>
> **Problem**
> Every day, restaurant owners must decide how much food to prepare and which ingredients to purchase without knowing exactly how much customers will order. Unexpected demand, unsold prepared food and ingredients expiring before use can lead to waste, lost revenue and lower profit margins. Small restaurants often lack affordable tools to analyze sales patterns and translate them into practical preparation and purchasing decisions.
>
> **Promise**
> Restaurant owners can make smarter daily preparation and purchasing decisions, reduce avoidable food waste, protect their margins and spend less time guessing how much food to prepare.
>
> **Why you**
> I started Cumin Kitchen, a food venture in Amravati, to understand business and entrepreneurship firsthand. Managing a food business helped me appreciate the challenges of running daily operations. As an aspiring restaurant owner, I want to use AI to solve practical problems faced by small food businesses like the one I hope to build.
>
> **Plan**
> I have identified food waste and demand uncertainty as problems worth investigating in small restaurants. Next, I will speak with 3-5 restaurant owners or kitchen managers in Bengaluru to understand their current practices, challenges and costs. Within three days, I plan to build an AI-powered prototype where users enter past sales, menu items and available inventory to receive suggested preparation quantities, purchasing recommendations and waste-risk alerts. I will test the prototype using sample restaurant data and refine it using feedback from potential users.

What was built

A single web page that runs in the browser with no sign-up, no hardware and no point-of-sale switch. The owner enters three things and gets a daily plan.

Input	Format
Past sales	One line per record: date, dish, quantity sold
Menu	Dish | price | ingredient:amount per portion
Inventory	Ingredient | unit | on hand | days until it expires | cost per unit

Output	What it tells the owner
What to prep	Expected sales per dish for the chosen day and suggested portions to prepare, including a safety buffer
What to buy	Ingredients needed for that prep, minus stock on hand, with the estimated cost of the shortfall
Waste-risk alerts	Stock that will probably not be used before it expires, with the rupee value at risk
Waste log	Records food thrown away with a reason (over-prepared, spoiled, plate waste, kitchen mistake, returned, other) and totals the cost by reason

How it works

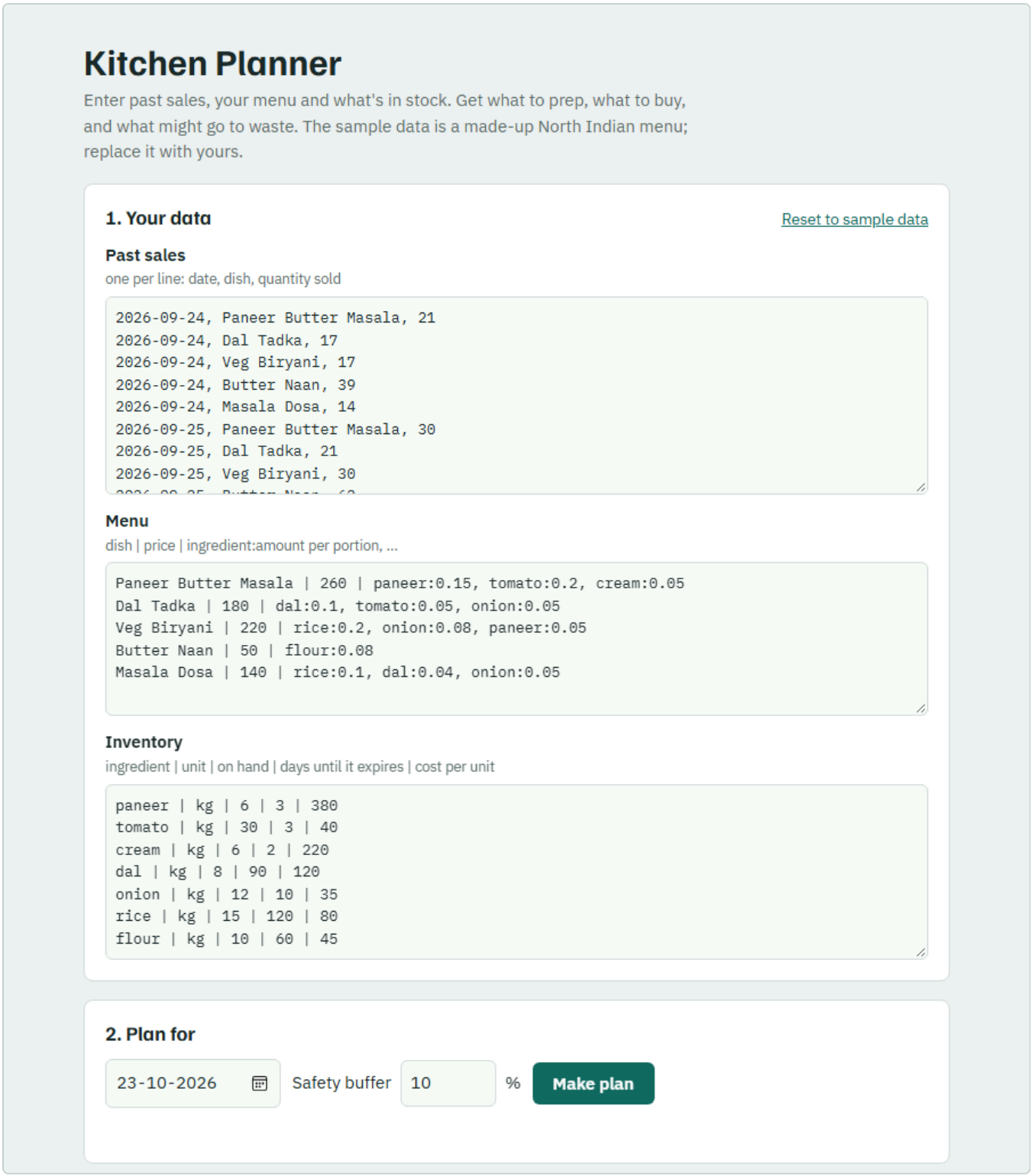
- **Demand forecast.** For each dish, a weighted average of past sales on the same weekday. Each week further back counts 30% less. Days with no record count as zero sales.
- **Prep quantity.** Forecast plus the safety buffer (default 10%), rounded up to whole portions.
- **Purchasing.** Recipe amounts multiplied by prep portions, minus current stock.
- **Waste risk.** Forecast usage from the plan date until each ingredient's expiry date (up to 14 days). If more than 20% of the stock would be left, an alert is raised. Alerts of 2 days or less are marked red.
- **Privacy.** All data stays in the user's browser. Nothing is sent to a server.

Market check

PetPooja is a restaurant POS with recipe-based stock deduction, low-stock alerts, purchase orders and wastage tracking. It records what was used and wasted. **Winnow** and **LeanPath** use smart scales and cameras to log food waste with reasons, and target larger kitchens. Kitchen Planner aims at **prevention**: it suggests what to prepare and buy before the waste happens, with no hardware. The waste-reason log borrows the idea from Winnow.

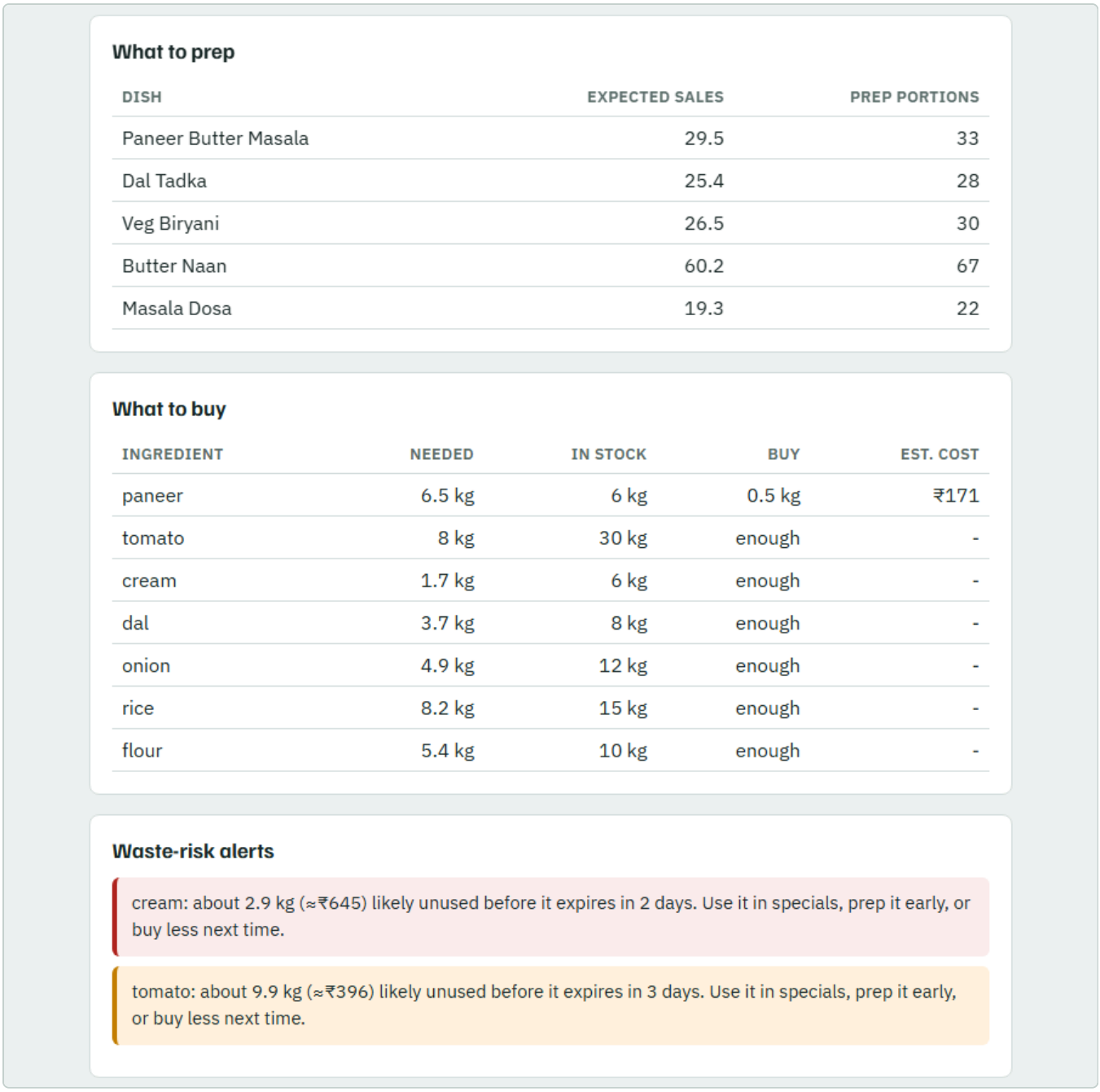
Findings from public product pages and review sites. Vendor claims such as percentage waste reductions were not independently verified.

Screenshot 1: data entry and plan date



Sample data is loaded on first open: 4 weeks of sales for a five-dish North Indian menu, its recipes, and seven ingredients in stock. The owner picks a date and a safety buffer, then clicks Make plan.

Screenshot 2: prep plan, purchase list and waste-risk alerts



What to prep

DISH	EXPECTED SALES	PREP PORTIONS
Paneer Butter Masala	29.5	33
Dal Tadka	25.4	28
Veg Biryani	26.5	30
Butter Naan	60.2	67
Masala Dosa	19.3	22

What to buy

INGREDIENT	NEEDED	IN STOCK	BUY	EST. COST
paneer	6.5 kg	6 kg	0.5 kg	₹171
tomato	8 kg	30 kg	enough	-
cream	1.7 kg	6 kg	enough	-
dal	3.7 kg	8 kg	enough	-
onion	4.9 kg	12 kg	enough	-
rice	8.2 kg	15 kg	enough	-
flour	5.4 kg	10 kg	enough	-

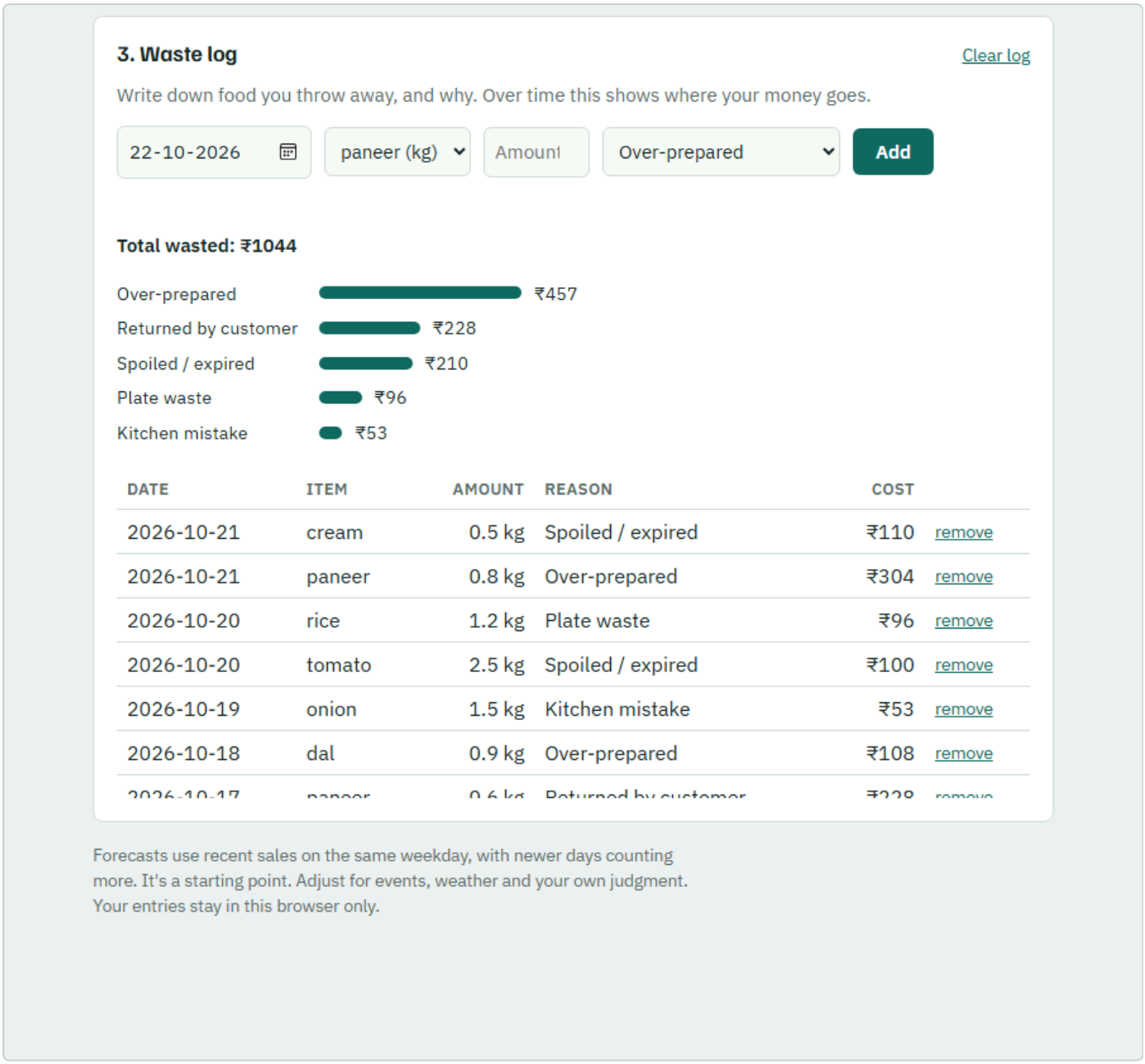
Waste-risk alerts

cream: about 2.9 kg (≈₹645) likely unused before it expires in 2 days. Use it in specials, prep it early, or buy less next time.

tomato: about 9.9 kg (≈₹396) likely unused before it expires in 3 days. Use it in specials, prep it early, or buy less next time.

For 23 October the planner suggests 33 paneer butter masala, 28 dal tadka, 30 veg biryani, 67 naan and 22 masala dosa. It says to buy 0.5 kg of paneer (about Rs 171). It also warns that about 2.9 kg of cream (about Rs 645) and 9.9 kg of tomato (about Rs 396) will probably expire unused, because the sample stock is deliberately higher than demand.

Screenshot 3: waste log with reason codes



Eight sample waste entries from the past week total Rs 1,044. "Over-prepared" is the largest cause (Rs 457), followed by food returned by customers (Rs 228) and spoilage (Rs 210). This tells an owner where to act first.

Testing so far

- Planning logic run on sample data in Node.js. Outputs checked by hand for sensible prep, purchase and alert values.
- Page rendered in Microsoft Edge to produce the screenshots above, with correct currency symbols.
- Not yet tested: phone screens, real restaurant data, or feedback from restaurant owners.

Limitations

- The forecast is a statistical method (weighted weekday average), not a machine-learning or language model. It does not know about festivals, weather, promotions or special events.
- All numbers on these screens come from made-up sample data, not a real restaurant.
- Data is typed or pasted in by hand. There is no direct import from POS systems yet.
- Data is stored only in one browser on one device.

Next steps

Step	Purpose
Interview 3-5 restaurant owners or kitchen managers in Bengaluru	Learn current prep and purchasing practice, waste costs, and which inputs owners would actually enter
Compare predicted and actual waste	Measure whether the planner's suggestions reduce waste
Import POS sales exports (CSV)	Remove manual data entry
AI-written plain-language suggestions	Explain the numbers in everyday words, for example which dish to put on special tonight
Test on a real restaurant's data	Check forecast accuracy against reality and tune the method

Links

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